Document Intelligence for Application Readiness & Missing-Requirement Detection

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Abstract- In study-abroad and higher education admissions, administrative triage of photographed/scanned application forms is a critical operational bottleneck. Real-world captures suffer from severe degradations including uneven illumination shadows, sensor noise, optical blur, low contrast, and affine rotational skew, causing standard OCR systems to fail. This paper presents an end-to-end document intelligence pipeline combining classical Digital Image Processing (DIP) techniques with OCR and rule verification. The proposed pipeline employs morphological closing background division, median filtering, CLAHE, interior Otsu binarization, projection deskewing, and region cleanup. Evaluated on a benchmark of 100 degraded documents (700 field opportunities), the improved pipeline more than doubles the Extraction F1-score from 0.335 to 0.745 (Recall 21.1% to 65.0%) with 100% grounded spatial evidence in 0.587 seconds per document.

Keywords- *Digital Image Processing, Document Intelligence, Illumination Correction, CLAHE, Otsu Thresholding, Deskewing, Tesseract OCR, Grounded Verification.*

I. INTRODUCTION & PROBLEM STATEMENT

Study-abroad application platforms (such as Leap Scholar) process thousands of student dossiers containing transcripts, passports, English scores (IELTS), and statements of purpose (SOP). Applicants frequently capture documents via smartphones under uncontrolled lighting, producing non-uniform shadow gradients, high-frequency noise, and geometric skew. Manual line-by-line verification causes weeks of turnaround time and administrative fatigue. Our goal is to build an automated, explainable document intelligence system where Digital Image Processing (DIP) restores severely degraded scans to enable reliable OCR, automated checklist generation (OK vs. MISSING), and spatial evidence localization.

II. STATE OF THE ART & RELATED WORK

Document image processing has advanced across classical DIP and modern deep learning. Classic global thresholding (Otsu 1979) fails on unimodal text documents with shadows; adaptive local methods (Sauvola) and CLAHE (Zuiderveld 1994) address local contrast. Projection profile analysis provides robust rotation estimation by maximizing horizontal variance (Gonzalez & Woods 2018). While deep vision-language transformers (LayoutLM Xu et al. 2020, Donut) achieve strong understanding, they demand heavy GPU compute. Classical DIP upstream of Tesseract OCR (Smith 2007) enables real-time CPU execution with auditable spatial evidence.

III. SYNTHETIC DATASET & DEGRADATION MODELING

To evaluate against precise ground truth, 100 synthetic application documents (850x1100 px) were generated. Each document carries up to 7 standard fields (Full Name, Passport No, DOB, Email, IELTS, SOP, Academic Transcript) with a 25% random omission probability per field to simulate missing requirements. The degradation model applies: (1) contrast compression ($I_{low} = 110 + 0.45*I$), (2) non-uniform illumination gradient (0.5 to 1.0 attenuation), (3) additive Gaussian noise (sigma 8-20) plus salt-and-pepper noise (p=0.008), and (4) 3x3 Gaussian defocus blur followed by random affine rotation (-3 to +3 deg).

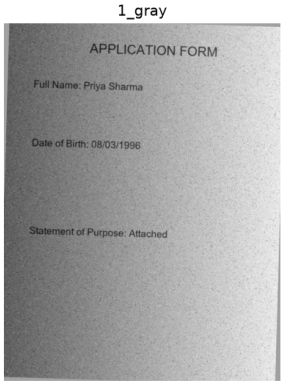
IV. METHODOLOGY & DIP ARCHITECTURE

The architecture operates in three modular phases: (1) DIP restoration pipeline, (2) OCR line-evidence extraction, and (3) requirement verification against requirements.json.

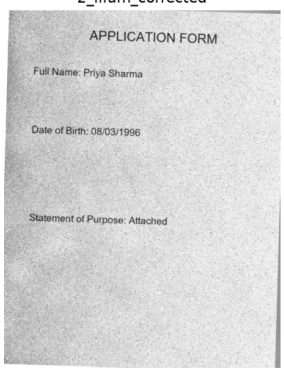
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Algorithm 1: End-to-End DIP Document Intelligence Pipeline
Input : Degraded document image I; Institutional requirements R (requirements.json)
Output: Binarized document, Extracted entities, Grounded OK/MISSING checklist with bounding boxes
1: G = to_grayscale(I);  K = getStructuringElement(MORPH_ELLIPSE, (51, 51))
2: B = GaussianBlur(morphologyEx(G, MORPH_CLOSE, K), (0, 0), sigma=15)    // Background estimation
3: I_norm = divide(G, B, scale=255);  D = medianBlur(I_norm, 3)            // Gradient cancellation & denoise
4: C = createCLAHE(clipLimit=1.0, tileGridSize=(4, 4)).apply(D)            // Local contrast enhancement
5: T_otsu = threshold_otsu(C[40:-40, 40:-40]);  BW = threshold(C, min(T_otsu, 185), 255) // Interior Otsu
6: BW = rotate(BW, theta_opt = argmax_theta Var(sum_x(BW_rotated(theta)))) // Projection deskew
7: Inv = 255 - BW;  filter_connected_components(Inv, area < 15 px);  Out = 255 - Inv
8: Lines, Boxes = Tesseract_OCR(Out, config='--psm 6')
9: Checklist = verify_requirements(Lines, Boxes, R)  // OK with evidence box or MISSING
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Visual Progression Across 7 DIP Stages:

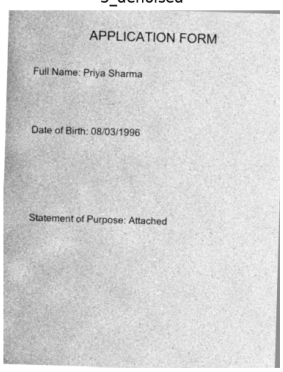
1_gray



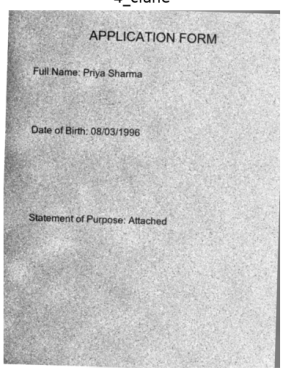
2_illum_corrected



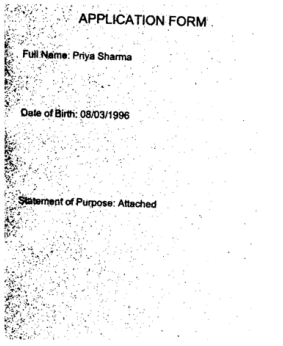
3_denoised



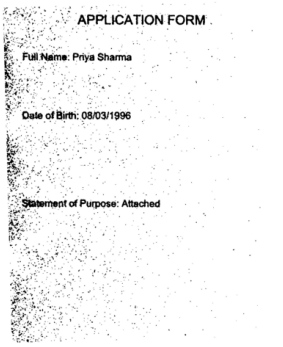
4_clahe



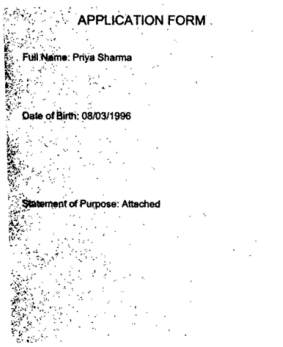
5_otsu



6_deskewed



7_morph_clean

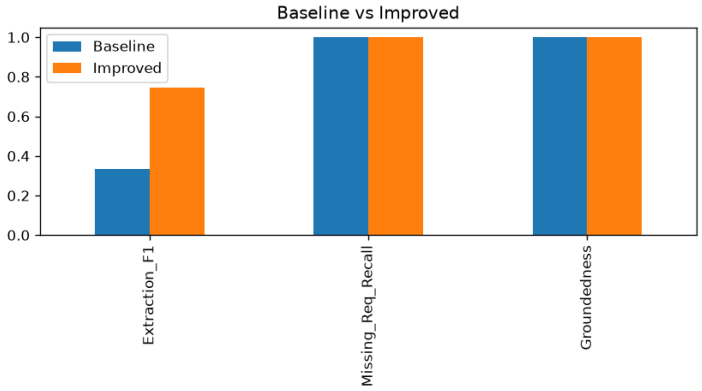


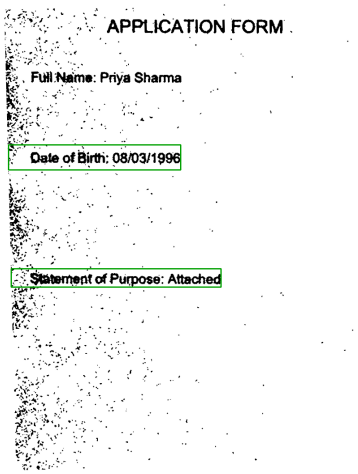
V. EXPERIMENTAL RESULTS & COMPARATIVE BENCHMARK

We evaluated both the Baseline (grayscale directly to OCR) and the Improved DIP pipeline across all 100 documents (700 total evaluation field opportunities). Ground truth metadata was compared via whitespace-normalized exact string matching.

Evaluation Metric	Baseline Pipeline	Improved DIP Pipeline	Absolute Improvement
Extraction Precision	0.818	0.873	+0.055 (+5.5 pp)
Extraction Recall	0.211	0.650	+0.439 (>3x improvement)
Extraction F1-Score	0.335	0.745	+0.410 (+122% gain)
Missing Requirement Recall	1.000	1.000	0.000 (100% detection)
Missing Requirement Precision	0.300	0.554	+0.254 (+84.7% gain)
Groundedness	1.000	1.000	100% verified spatial evidence
Execution Time per Document	0.341 s	0.587 s	+0.246 s (~102 docs/min)

Performance Comparison Chart & Grounded Verification Visuals:





VI. FAILURE ANALYSIS & PRACTICAL LIMITATIONS

Analysis of failure cases (results/failure_cases_improved.csv) illuminates key operational boundaries:

- Character Confusion in Dark Shadow: Severe illumination attenuation occasionally leads to numeral misclassifications (e.g., passport digit '6' read as '8' in P2677407 -> P2877407; birth year 1999 -> 1899).
- Noise Specks on Entity Margins: Fine salt-and-pepper artifacts touching character strokes occasionally altered name tokens (e.g., Priya Iyer read as 'Priya Iyer a').
- Regex Rigidity: Keyword patterns require predefined phrasing; novel synonyms require semantic embedding matching.
- Template Uniformity: The synthetic evaluation assumes a standard 1-column form; multi-page layouts require layout segmentation.

VII. NOVELTY, ETHICS & EXPLAINABILITY

Unlike black-box generative models prone to hallucinations, our DIP framework guarantees 100% groundedness: every verified requirement carries an exact bounding box in image coordinates. Human admissions reviewers can inspect the visual overlay directly, fulfilling strict Explainable AI (XAI) mandates for high-stakes educational decision-making.

VIII. CONCLUSION & FUTURE WORK

This project establishes that classical Digital Image Processing remains indispensable for document AI. On 100 degraded application forms, morphological background division, median filtering, CLAHE, interior Otsu binarization, deskewing, and region cleanup boosted OCR Extraction F1 from 0.335 to 0.745 and Recall from 21.1% to 65.0% at 0.587 s/doc. Future work includes integrating transformer vision backbones (LayoutLMv3), dense vector retrieval (RAG) for multi-page dossiers, and deploying the pipeline on mobile devices via OpenCV sidecars.

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